

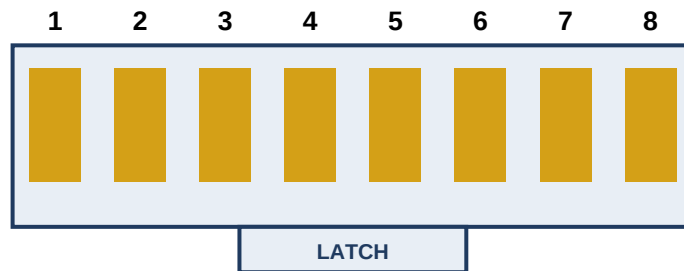
Ethernet Standards and Specifications

Answers to the practice questions

Q1. RJ-45 pin layout with T568A and T568B colour codes

Draw a simple diagram showing the pin layout of an RJ-45 connector and label the colour codes for T568A and T568B.

RJ-45 Connector - Front View (Pin 1 on left, latch down)



T568A

Pin	Colour	Wire
1		White/Green
2		Green
3		White/Orange
4		Blue
5		White/Blue
6		Orange
7		White/Brown
8		Brown

T568B

Pin	Colour	Wire
1		White/Orange
2		Orange
3		White/Green
4		Blue
5		White/Blue
6		Green
7		White/Brown
8		Brown

Striped box = white/colour wire. Only the green and orange pairs swap between A and B (pins 1,2,3,6).

Key points: Hold the plug with the contacts facing you and the latch pointing down; pin 1 is on the left. T568A and T568B differ only by swapping the green and orange pairs. Pins 1, 2 (transmit) and 3, 6 (receive) are used by 10/100 Mbps Ethernet; all four pairs are used by Gigabit Ethernet.

Q2. Comparison of 10BASE-T, 100BASE-TX and 1000BASE-T

Create a table comparing the key specifications of the three Ethernet standards.

Specification	10BASE-T	100BASE-TX	1000BASE-T
IEEE standard	802.3i	802.3u	802.3ab
Common name	Ethernet	Fast Ethernet	Gigabit Ethernet
Speed	10 Mbps	100 Mbps	1000 Mbps (1 Gbps)

Specification	10BASE-T	100BASE-TX	1000BASE-T
Cable type	UTP, Cat 3 or better	UTP, Cat 5 or better	UTP, Cat 5e or better (Cat 6 recommended)
Pairs used	2 pairs (1,2 and 3,6)	2 pairs (1,2 and 3,6)	4 pairs (all 8 pins, bidirectional)
Max. cable length	100 m	100 m	100 m
Connector type	RJ-45 (8P8C)	RJ-45 (8P8C)	RJ-45 (8P8C)
Line encoding	Manchester	4B5B + MLT-3	PAM-5

Q3. Connecting two laptops directly

Which cable would you use, straight-through or crossover, and why?

Answer: a crossover cable.

A laptop NIC is a DTE (end device): it transmits on pins 1 and 2 and receives on pins 3 and 6. If two laptops are joined with a straight-through cable (pin 1 to pin 1, pin 3 to pin 3, and so on), the transmit pins of laptop A reach the *transmit* pins of laptop B, so neither side listens to the other and no link forms.

A crossover cable swaps the pairs: pin 1 to pin 3 and pin 2 to pin 6 (and pin 4/5 to pin 7/8 for Gigabit). Now laptop A's transmit pair lands on laptop B's receive pair and vice versa, so communication works.

Cable	Wiring	Used to connect
Straight-through	Same standard at both ends (A-A or B-B)	Unlike devices: PC to switch, router to switch
Crossover	T568A at one end, T568B at the other	Like devices: PC to PC, switch to switch

Note: Most modern laptops support **Auto-MDI/MDI-X**, which swaps the pairs internally, so even a straight-through cable usually works today. Gigabit Ethernet (1000BASE-T) includes this feature as part of the standard, but the crossover cable is the correct answer by the basic theory.

Q4. Real-world devices that use Ethernet

List three devices in your home or college that use Ethernet, with the likely Ethernet standard.

Device	Where used	Likely standard	Reason
Desktop PC / laptop	Computer lab, hostel, home	1000BASE-T	Modern NICs ship with Gigabit ports for fast file transfer and internet access.
Wi-Fi router / access point / switch	Home, classroom, campus network	1000BASE-T (LAN ports)	LAN ports are Gigabit so wired and wireless users get full broadband speed.
Network printer	Office, library, college lab	100BASE-TX (older) or 1000BASE-T	Print jobs need little bandwidth, so Fast Ethernet is often enough.